CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Task 5: Quality attribute scenarios

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Team	RapidMed
Course	CSE4203 Principles of Software Architecture
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5. Quality attribute scenarios

QAS-1. Availability: no data lost offline

Part	Description
Source	Field paramedic
Stimulus	Records vital signs and a triage level while the ambulance computer has no network connection
Artifact	Paramedic app and its local store
Environment	No connection for up to 30 minutes; the app is closed and reopened during the outage
Response	Each entry is saved on the device before the app confirms it; entries are queued and sent in order when the connection returns; the server ignores any entry it has already received
Response measure	In 20 test outages of 1 to 30 minutes, with 500 entries in total and 5 forced app closures: no entry lost, no entry duplicated on the server, and every queued entry stored on the server within 30 s of the connection returning

QAS-2. Availability: routing service failure

Part	Description
Source	Google Routes API (external system)
Stimulus	Stops answering: every request times out or returns an error
Artifact	Routing adapter of the dispatch and routing engine
Environment	Normal operation, 20 destination requests per minute
Response	Calls time out after 1.5 s; after 3 consecutive failures a circuit breaker opens, and the engine ranks hospitals using travel times cached in the last 30 minutes, or straight-line distance where none is cached, and labels the result “estimated”; it tries the service again every 30 s and closes the breaker when a call succeeds
Response measure	During a 10-minute simulated outage, 100% of destination requests return a ranked result labelled “estimated” within 3 s; once the breaker is open, no request waits for the service; normal ranking returns within 60 s of the service recovering

QAS-3. Performance: destination at peak load

Part	Description
Source	100 simulated ambulance crews
Stimulus	Request a destination: 200 requests within 60 s
Artifact	Dispatch and routing engine
Environment	Peak load; 25 candidate hospitals; the Routes API is replaced by a mock that answers in 300 ms
Response	Ranks the hospitals by free beds of the type needed and by travel time, and returns the top three with an offer to reserve
Response measure	95% of requests answered within 3 s and 99% within 5 s, measured at the app, with no errors

QAS-4. Consistency (safety): the last bed

Part	Description
Source	Two field paramedics
Stimulus	Both request the last free resuscitation bed at the same hospital within 100 ms; one request is retried after a network time-out
Artifact	Reservation component of the dispatch and routing engine, and its database
Environment	Normal operation
Response	Exactly one reservation succeeds; the retry returns the same result instead of creating a second reservation; the other crew is told at once and offered the next-best hospital
Response measure	Over 1,000 runs, 0 double reservations and 0 extra reservations from retries; the losing crew sees an alternative within 3 s in 95% of runs

QAS-5. Security: break-glass access

Part	Description
Source	An authenticated paramedic treating an unconscious patient
Stimulus	Opens the patient’s summary through break-glass, giving a reason
Artifact	Break-glass path and audit log of the dispatch and routing engine
Environment	Normal operation; the patient cannot consent
Response	The summary opens; an audit record of the user, ambulance, patient, reason and time is written to an append-only log and flagged for review
Response measure	The summary opens within 3 s in 95% of 200 test requests; 100% of them have a complete audit record within 5 s; 0 of 50 requests without a reason succeed; 0 of 20 attempts to edit or delete an audit record succeed from any application role; one month’s break-glass accesses are listed in under 1 minute

QAS-6. Interoperability: Sanad under a rate limit

Part	Description
Source	15 ambulance crews
Stimulus	90 patient summary requests within 60 s
Artifact	Sanad FHIR client of the dispatch and routing engine
Environment	Peak load; Sanad’s limit is assumed to be 60 requests per minute
Response	Sends standard HL7 FHIR R4 requests carrying RapidMed’s registered client credentials, the user and the reason; sends at most 60 per minute and queues the rest; answers from the episode cache when a summary was already fetched
Response measure	In a 10-minute test, 0 requests above 60 per minute and 0 rate-limit errors from the Sanad mock; 100% of 200 sampled requests are valid FHIR R4 and carry user and reason; 95% of the first 60 requests are answered within 3 s and the other 30 within 60 s; the mock rejects all 50 requests sent without valid client credentials

QAS-7. Usability: recording vital signs

Part	Description
Source	A field paramedic who has not used RapidMed before
Stimulus	Records heart rate, blood pressure, respiratory rate, oxygen saturation and a triage level
Artifact	Vital-signs screen of the paramedic app
Environment	Normal operation; the paramedic wears examination gloves; the computer is mounted as in the vehicle
Response	One “Record vitals” action opens a single screen with large fields and number pads; implausible values are flagged before saving
Response measure	In a test with 10 first-time users, at least 8 record the full set within 60 s, with no more than 6 taps per vital sign and no uncorrected error; the System Usability Scale score averages at least 68

QAS-8. Modifiability: replacing the routing provider

Part	Description
Source	Development team
Stimulus	Replace the Google Routes API with another routing provider, for example because the free allowance runs out
Artifact	Routing adapter of the dispatch and routing engine
Environment	Development, after the prototype
Response	The new provider is implemented behind the existing routing interface; nothing outside the adapter changes
Response measure	Only files in the routing adapter and its configuration change, as checked in the pull request; the work takes no more than 2 person-days; all existing routing tests pass against the new adapter